

- Decimal Number System
- Binary Number System
- Binary $\rightleftharpoons$ Decimal
- Addition of Binary Nos.
- Bitwise Operators
- Binary Representation of -ve nos.
- Range of data types
- Importance of constraints

Intermediate Contest



- ① Date : 17 Nov
- ② 9 - 10:30 PM (1.5 hrs ; 3 Ques)
- ③ Syllabus : Intermediate Module
- ④ Discussion at 10:30 PM same day

Decimal Number System $\begin{matrix} \rightarrow \text{base } 10 \\ \rightarrow \text{digits: } 0-9 \end{matrix}$

$$\begin{array}{r} 2 \quad 1 \quad 0 \\ 3 \quad 4 \quad 2 \end{array} = 300 + 40 + 2$$
$$= 3 \times 10^2 + 4 \times 10^1 + 2 \times 10^0$$

Place value of each digit = digit $\times 10^{\text{pos}}$

$$\begin{array}{r} 3 \quad 2 \quad 1 \quad 0 \\ 2 \quad 5 \quad 6 \quad 3 \end{array} = 2000 + 500 + 60 + 3$$
$$= 2 \times 10^3 + 5 \times 10^2 + 6 \times 10^1 + 3 \times 10^0$$

Binary Number System $\begin{matrix} \rightarrow \text{base } 2 \\ \rightarrow \text{Digits: } 0, 1 \\ \text{(bits)} \end{matrix}$

$$\begin{array}{r} 2 \quad 1 \quad 0 \\ 1 \quad 1 \quad 0 \end{array} = 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$$
$$= 4 + 2 + 0 = 6$$

Place value of each digit = digit $\times 2^{\text{pos}}$

$$(110)_2 = (6)_{10}$$

$$\begin{array}{r} 3 \quad 2 \quad 1 \quad 0 \\ 1 \quad 0 \quad 1 \quad 1 \end{array} = 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$
$$= 8 + 0 + 2 + 1$$
$$= 11$$

$$(1011)_2 = (11)_{10}$$

Binary to Decimal

* Multiply each digit by 2^{pos}

* Add up to get result

$$\begin{array}{cccccc} 4 & 3 & 2 & 1 & 0 & \\ 1 & 0 & 1 & 0 & 1 & \\ \downarrow & & \downarrow & & \downarrow & \\ 1 \times 2^4 & & 1 \times 2^2 & & 1 \times 2^0 & = 1 + \\ & & & & & 4 + \\ & & & & & \underline{16} \\ & & & & & 21 \end{array}$$

$(10101)_2 = (21)_{10}$

$$\begin{array}{ccccccc} 6 & 5 & 4 & 3 & 2 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ \downarrow & & \downarrow & \downarrow & & \downarrow & \\ 2^6 & & 2^4 & 2^3 & & 2^1 & = 90 \\ \downarrow & & \downarrow & \downarrow & & \downarrow & \\ 64 & + & 16 & + 8 & + & 2 & \end{array}$$

$(1011010)_2 = (90)_{10}$

1 $\rightarrow$ set bit , 0 $\rightarrow$ unset bit

Decimal to Binary

- ① Keep Dividing the no. by 2 till the no. becomes 0
- ② Collect remainders from bottom to top

2	20	-	0
2	10	-	0
2	5	-	1
2	2	-	0
2	1	-	1
	0		

$$(20)_{10} \rightarrow (10100)_2$$

2	45	-	1
2	22	-	0
2	11	-	1
2	5	-	1
2	2	-	0
2	1	-	1
	0		

$$(45)_{10} = (101101)_2$$

Addition of Decimal Nos.

$$\begin{array}{r}
 \textcircled{1} \quad \textcircled{1} \\
 3 \quad 6 \quad 8 \\
 + \quad 4 \quad 5 \quad 3 \\
 \hline
 8 \quad 2 \quad 1
 \end{array}$$

→ Start from right most digit

→ sum of digits = $8 + 3 = 11$

$\begin{array}{c} c \quad d \\ \swarrow \quad \searrow \\ 1 \quad 1 \end{array}$

→ sum of digits = $1 + 6 + 5 = 12$

$\begin{array}{c} c \quad d \\ \swarrow \quad \searrow \\ 1 \quad 2 \end{array}$
 $\text{sum}/10 \quad \text{sum} \cdot 10$

→ sum of digits = $1 + 3 + 4 = 8$

$\begin{array}{c} c \quad d \\ \swarrow \quad \searrow \\ 8/10 = 0 \quad 8 \cdot 10 = 8 \end{array}$

Addition of Binary Nos.

Ex 1

$$\begin{array}{r}
 \textcircled{1} \textcircled{1} \textcircled{1} \textcircled{0} \textcircled{1} \\
 1 \quad 0 \quad 1 \quad 0 \quad 1 \\
 + \quad 0 \quad 1 \quad 1 \quad 0 \quad 1 \\
 \hline
 1 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0
 \end{array}
 \begin{array}{l}
 \rightarrow 21 \\
 + 13 \\
 \hline
 34
 \end{array}$$

sum

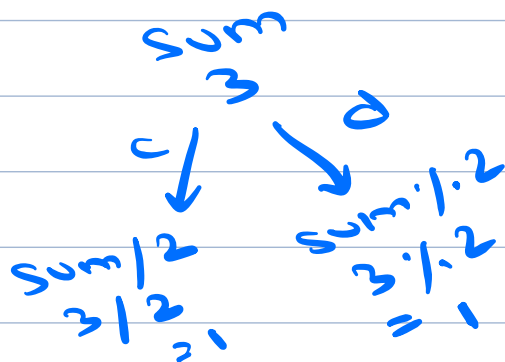
$\begin{array}{c} c \quad d \\ \swarrow \quad \searrow \\ 1/2 = 0 \quad 1 \cdot 2 = 1 \end{array}$

sum

$\begin{array}{c} c \quad d \\ \swarrow \quad \searrow \\ \text{sum}/2 \quad \text{sum} \cdot 2 \\ 2/2 = 1 \quad 2 \cdot 2 = 0 \end{array}$

A handwritten example of binary addition on lined paper. The first number, 1010101, is written in blue ink above the second number, 1101101, which is written in blue ink. A green horizontal line is drawn between the two numbers. Below the line, the sum 10001000 is written in blue ink. A red vertical mark is placed under the first column (the leftmost column) of the sum. A second green horizontal line is drawn below the sum.

Quiz :

$$\begin{array}{r} 01010 \\ + 00111 \\ \hline 11101 \end{array}$$


Bitwise Operators



- perform operations on bits of binary nos.
- $0 \rightarrow \text{false/unset}$ $1 \rightarrow \text{true/set}$
- used to manipulate binary data

A	B	$A \& B$	$A B$	$A \wedge B$	$\sim A$
		AND (&)	OR ()	XOR (^)	NOT (!/~)
0	0	0	0	0	1
0	1	0	1	1	1
1	0	0	1	1	0
1	1	1	1	0	0

XOR $\rightarrow$ same bits, XOR = 0
diff bits, XOR = 1

NOT $\rightarrow$ Reverse / Toggle / swap the bit

XOR



same same
puppy shame

① $5 \& 6 = 4$

$5 \rightarrow 101$

$2 \quad 6 \rightarrow \begin{array}{r} 110 \\ \underline{100} \end{array}$

$\begin{array}{r} 2^1 \quad 0 \\ 100 \\ \downarrow \\ 2^2 = 4 \end{array}$

② $20 \& 45 = 4$

$20 \rightarrow 010100$

$2 \quad 45 \rightarrow \begin{array}{r} 101101 \\ \underline{000100} \\ \hline 001000 \end{array}$

③ $92 \mid 154 = 222$

$$\begin{array}{r}
 92 \rightarrow 01011100 \\
 154 \rightarrow 10011010 \\
 \hline
 11011110 \\
 \hline
 \begin{array}{ccccccc}
 7 & 6 & 5 & 4 & 3 & 2 & 1 & 0 \\
 2^7 & 2^6 & & 2^4 & 2^3 & 2^2 & 2^1 &
 \end{array}
 \end{array}$$

④ $92 \wedge 154 = 198$

$$\begin{array}{r}
 92 \rightarrow 01011100 \\
 \wedge 154 \rightarrow 10011010 \\
 \hline
 11000110 \\
 \hline
 \begin{array}{ccccccc}
 7 & 6 & 5 & 4 & 3 & 2 & 1 & 0 \\
 2^7 & 2^6 & & & & 2^2 & 2^1 &
 \end{array}
 \end{array}$$

⑤ $\sim 92 = 163$

$$\begin{array}{l}
 92 \rightarrow \sim (01011100) \\
 = (10100011) \rightarrow 163
 \end{array}$$

$$4 \rightarrow 100 \rightarrow 00000100$$

★ 8 bit binary representation

Quiz : $20 \wedge 45 = 57$

$$\begin{array}{r}
 20 \rightarrow 010100 \\
 \wedge 45 \rightarrow 101101 \\
 \hline
 111001 \\
 \hline
 \begin{array}{ccccccc}
 5 & 4 & 3 & 2 & 1 & 0 &
 \end{array} \\
 2^5 & 2^4 & 2^3 & & & 2^0 &
 \end{array}$$

Binary Representation of Negative Numbers

1. Convert absolute value of number to binary
2. Swap all bits of binary no. (1s complement)
3. Add 1 to no. obtained in step 2 (2s complement)

Convert -5 to its 8 bit binary representation

$$\begin{array}{rcl} \textcircled{1} & 5 \rightarrow & 0000101 \\ \textcircled{2} & & 1111010 \text{ (1s complement)} \\ \textcircled{3} & + & \underline{\quad\quad\quad 1} \\ & & \underline{1111011} \text{ (2s complement)} \end{array}$$

$$(-5)_{10} = (1111011)_2$$

$$\begin{array}{ccccccc} 7 & 6 & 5 & 4 & 3 & 2 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 & 0 & 1 & 1 \\ \downarrow & & \downarrow & \downarrow & \downarrow & & \downarrow & \downarrow \\ 2^7 & & 2^5 & 2^4 & 2^3 & & 2^1 & 2^0 \\ & \downarrow & & & & & & \\ & 2^6 & & & & & & \end{array} = 251$$

$$\begin{array}{cccccccc} \text{MSB} & & & & & & & \text{LSB} \\ \hline 7 & 6 & 5 & 4 & 3 & 2 & 1 & 0 \end{array}$$

Note - In case of signed numbers (+ve and -ve nos.), the MSB has a -ve base value

Left Most Bit $\rightarrow$ Most significant Bit (MSB)

Right Most Bit $\rightarrow$ LSB

$$\begin{array}{cccccccc}
 7 & 6 & 5 & 4 & 3 & 2 & 1 & 0 \\
 -2^7 & 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \\
 \hline
 \text{MSB} & & & & & & &
 \end{array}$$

$$\begin{array}{cccccccc}
 7 & 6 & 5 & 4 & 3 & 2 & 1 & 0 \\
 1 & 1 & 1 & 1 & 1 & 0 & 1 & 1 \\
 \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\
 -2^7 & 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \\
 \swarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\
 -128 & + & 64 & + & 32 & + & 16 & + & 8 & + & 2 & + & 1 \\
 \hline
 -128 & + & 123 & = & -5
 \end{array}$$

$$\begin{array}{cccccccc}
 7 & 6 & 5 & 4 & 3 & 2 & 1 & 0 \\
 -2^7 & 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0
 \end{array}$$

$$\begin{array}{cccccccc}
 7 & 6 & 5 & 4 & 3 & 2 & 1 & 0 \\
 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\
 -2^7 & 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0
 \end{array}$$

$$\begin{array}{cccccccc}
 7 & 6 & 5 & 4 & 3 & 2 & 1 & 0 \\
 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\
 -2^7 & 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0
 \end{array}$$

$$-128 + 64 + 32 + 16 + 8 + 4 + 2 + 1$$

$$-128 + 127 = -1$$

MSB $\rightarrow$ signed bit

0 $\rightarrow$ +ve
1 $\rightarrow$ -ve

10 : 35

Signed Int	Unsigned Int
MSB $\rightarrow$ sign bit	MSB $\rightarrow$
-ve base value	+ve base value

Convert -3 to its 8 bit binary representation

$$\begin{array}{rcl} \textcircled{1} & 3 & \rightarrow 00000011 \\ \textcircled{2} & & 11111100 \text{ (1s complement)} \\ \textcircled{3} & + & \underline{\quad\quad\quad 1} \\ & & \underline{11111101} \text{ (2s complement)} \end{array}$$

$$(-3)_{10} = (11111101)_2$$

	7	6	5	4	3	2	1	0	
	1	1	1	1	1	1	0	1	
-2^7	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	
	2^6	2^5		2^4	2^3	2^2	2^1	2^0	

$-128 + 64 + 32 + 16 + 8 + 4 + 1$
 $-128 + 125 = -3$

Convert -10 to its 8 bit binary representation

$$\begin{array}{rcl} \textcircled{1} & 10 & \rightarrow 00001010 \\ \textcircled{2} & & 11110101 \text{ (1s complement)} \\ \textcircled{3} & + & \underline{\quad\quad\quad 1} \\ & & \underline{11110110} \text{ (2s complement)} \end{array}$$

$$(-10)_{10} = (11110110)_2$$

Range of data types

Bits	Min	Max
4	$\begin{array}{cccc} 3 & 2 & 1 & 0 \\ \underline{1} & \underline{0} & \underline{0} & \underline{0} \\ -2^3 & 2^2 & 2^1 & 2^0 \\ = -2^3 = -8 \end{array}$	$\begin{array}{cccc} 3 & 2 & 1 & 0 \\ \underline{0} & \underline{1} & \underline{1} & \underline{1} \\ -2^3 & 2^2 & 2^1 & 2^0 \\ 2^2 + 2^1 + 2^0 \\ = 4 + 2 + 1 = 7 \end{array}$
N	$\begin{array}{ccccccc} N-1 & N-2 & & & 2 & 1 & 0 \\ \underline{1} & \underline{0} & \underline{0} & \dots & \underline{0} & \underline{0} & \underline{0} \\ -2^{N-1} & 2^{N-2} & & & 2^2 & 2^1 & 2^0 \\ -2^{N-1} \end{array}$	$\begin{array}{ccccccc} N-1 & N-2 & & & 2 & 1 & 0 \\ \underline{0} & \underline{1} & \underline{1} & \dots & \underline{1} & \underline{1} & \underline{1} \\ -2^{N-1} & 2^{N-2} & & & 2^2 & 2^1 & 2^0 \\ 2^{N-1} - 1 \end{array}$

$$[-2^{N-1}, 2^{N-1} - 1]$$

$$\text{Max no} = 2^0 + 2^1 + 2^2 + \dots + 2^{N-2}$$

AP $a=1, r=2, n=N-1$

$$\text{Sum} = \frac{a(r^n - 1)}{r - 1}$$

$$S = \frac{1(2^{N-1} - 1)}{2 - 1} = 2^{N-1} - 1$$

int : 4 B
↓
32 bits

$$[-2^{31}, 2^{31} - 1]$$

$$[-2 \times 10^9, 2 \times 10^9 - 1] \quad \begin{array}{l} 1024 \approx 10^3 \\ 2^{10} \approx 10^3 \\ 2^{30} \approx 10^9 \end{array}$$

int $\rightarrow [-10^9, 10^9]$

$$2^{31} = 2^{30} \times 2$$

long: 8 B

↓
64 bits

$$[-2^{63}, 2^{63} - 1]$$



$$[-8 \times 10^{18}, 8 \times 10^{18} - 1]$$

$$\text{long} \sim [-10^{18}, 10^{18}]$$

$$2^{63} = 2^3 \times 2^{60} = 8 \times 10^{18}$$

$$2^{60} \approx 10^{18}$$

16 in 4 bits?

1 0 0 0 0

garbage value = 0

When you try to store a bigger no. into a smaller data type, overflow happens. You get garbage value

128 →
(8 bits)

7 6 5 4 3 2 1 0
1 0 0 0 0 0 0 0
2⁷ 2⁶ 2⁵ 2⁴ 2³ 2² 2¹ 2⁰

8 bits
-128 to 127

17 in 4 bits

→ 1 0 0 0 1

garbage value = 1

Importance of constraints

int a = 10^5

int b = 10^6

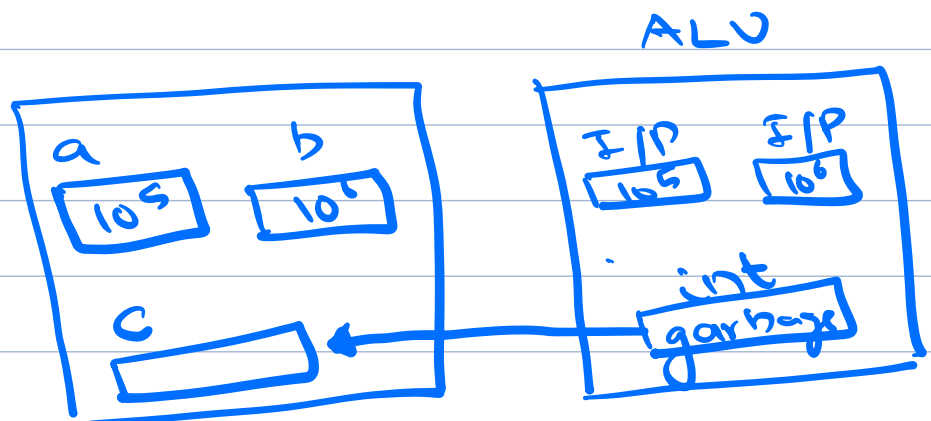
TRY 1

int c = a * b

// $10^5 \times 10^6 = 10^{11}$

It will overflow as 10^{11} is large for integer

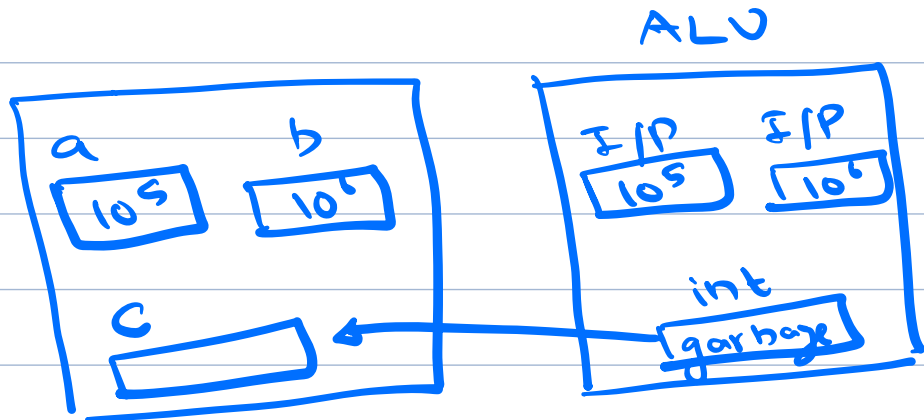
- ① The calculation happens at ALU
- ② If we provide ALU with 2 int, it calculates result in INT



16 (10000) $\xrightarrow{4 \text{ bits}}$ 0000 $\xrightarrow{8 \text{ bits}}$ 00000000

TRY 2

$\text{long } c = a * b$ X



TRY 3

$\text{long } c = \text{long}(a * b)$ No

It still overflows

- ① $a * b \rightarrow$ ans garbage
- ② Typecast ans to long
- ③ Store it in c

TRY 4

$\text{long } c = (\text{long}) a * b$ ✓

- ① a will be typecasted to long

long int

- ② $a * b \rightarrow \text{Long}$

- ③ Product will get it copied to c

I/P I/P
long int

long

$$\text{long } c = a * (\text{long}) b$$

Ques. $1 \leq N \leq 10^5$

$$1 \leq A[i] \leq 10^6$$

Find sum of array

~~long~~

~~int~~ sum = 0

for ($i=0$; $i < n$; $i++$) <

 sum = sum + A[i]

print (sum)

Max value of sum = $10^5 \times 10^6 = 10^{11}$
